Acme Logistics Corp: Technical RFP & Project Brief

Company Background & Problem Statement

Acme Logistics manages a fleet of over 500 regional delivery vehicles in the GCC. Currently, our dispatch team relies on chaotic internal emails and manually filled excel files to schedule driver shifts and log customer delays. This causes massive coordination friction and operational blind spots.

Target Architecture Objectives

We want to build a centralized intelligent dispatch portal. We require a web-based dashboard for our operations control staff and a sleek mobile web layout for delivery drivers on the road. The backend needs to run on robust cloud foundations and communicate efficiently with legacy systems.

Core Feature Scope Requirements

1. Automated Shift Planning: An AI scheduler engine that reads historical driver logs and auto-suggests optimal weekly shift rosters.
2. Multi-Channel Alert Router: An automated messaging worker that updates customers on delivery delays instantly using WhatsApp APIs and SMS fallbacks.
3. Live Analytics View: A dashboard using responsive charts to monitor driver performance metrics, idle times, and active trip maps.

Technical Parameters & Infrastructure Expectations

The software architecture must utilize Python for data workflows and core processing engines. The administrative web dashboard needs to be interactive, using something modern like React. We want to avoid a massive infrastructure layout from day one, so we are open to leveraging secure third-party APIs (like Twilio or Make) to handle multi-channel alerts and communication workflows quickly.